An Improved Compound Gaussian Model for Bivariate Surface EMG Signals Related to Strength Training

Durgesh Kusuru, *Student Member, IEEE*, Anish C. Turlapaty, *Senior Member, IEEE*, and Mainak Thakur, *Member, IEEE*

***Abstract*—Recent literature suggests that the surface electromyography (sEMG) signals have nonstationary statistical characteristics, specifically due to the random nature of the covariance. Thus, the suitability of a statistical model for sEMG signals is determined by the choice of an appropriate model for describing the covariance. The purpose of this study is to propose a compound-Gaussian (CG) model for multivariate sEMG signals in which the latent variable of covariance is modeled as a random variable that follows an exponential model. The parameters of the model are estimated using the iterative expectation maximization (EM) algorithm. Further, a new dataset, electromyography analysis of human activities database 2 (EMAHA-DB2), is developed. The proposed model is evaluated through both qualitative and quantitative methods. Based on the model fitting analysis on the sEMG signals from EMAHA-DB2, it is found that the proposed CG model fits more closely to the empirical pdf of sEMG signals than the existing models. In addition, statistical analyses are carried out among the models and estimated parameters under different scenarios. The estimate of the exponential model's rate parameter exhibits a clear relationship with training weights, potentially correlating with underlying motor unit activity. Finally, the average signal power estimates of the channels show distinctive dependency on the training weights, the subject's training experience, and the type of activity.**

***Index Terms*—Compound Gaussian (CG) models, expectation maximization (EM) algorithm, exponential random variable, surface electromyography (sEMG).**

I. Introduction

A. Statistical Models of sEMG Signals

STATISTICAL models of strength of surface electromyography (sEMG) signals have many applications, such as 1) to enhance the interpretation of the sEMG signals in clinical studies including detection of neuromuscular disorders [1]; 2) to build visualization tools to support movement sciences [2], and muscle physiology examinations, and the sports science education; 3) to improve system identification models that noninvasively determine the muscle force and the joint torque [3]; 4) to understand interrelationships between sEMG signals and muscle groups, for example in sports activities [4], [5], [6]; 5) to develop insights into sEMG signal generation from the constituent motor unit action potentials (MUAPs) that form basis for the sEMG signal synthesis [7] and simulation studies [8]; and 6) to improve performance for pattern classification of intended actions to control wearable exoskeletons and prostheses [9]. The sEMG signals can be modeled as stochastic processes because each constituent motor unit firing can be considered a random event [10]. The shape of the probability density function (pdf) of EMG signals depends on many factors, such as muscle contraction force level, the type of muscle, and noise. Furthermore, the distribution may also be influenced by the nature of the activity (static or dynamic), the subject's position during the activity (sitting, standing, or running), and the choice of limb (upper or lower), etc. [11]. In one of the earliest experiments [12], the Gaussian distribution was used to explain the statistical nature of EMG signals. A study by Brown and Stein [13] showed that in the presence of constant-force conditions, the distribution of EMG signals collected from the bicep and first dorsal interosseous muscles underwent a sharper peak than that of the Gaussian distribution. A few more studies [3], [14], [15], [16] show that the nonGaussianity of EMG signals varies according to the level of muscle contraction; for example, as it increases, the distribution of EMG signals shifts toward the Gaussian. In contrast, Hussain et al. [11] suggested that the distribution tends toward the nonGaussian when the level of contraction force increases. In summary, in the earlier literature [3], [11], [16], sEMG signals were considered stationary signals following unimodal distributions.

B. Motivation for Compound Models

In Furui et al. [10], a motivation for modeling the EMG signal as a compound random variable is presented as follows: Based on [17], the muscle force estimation can be related to variance estimation under Gaussian assumptions. However, Furui et al. [10], reasoned that the constant variance model in [17] is

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The authors are with the Bio-Signal Analysis Group, Indian Institute of Information Technology Sri City, Chittoor, Andhra Pradesh 517646, India (e-mail: durgesh.k@iiits.in; anish.turlapaty@iiits.in; mainak.thakur@iiits.in).

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not sufficient based on evidence from the following studies [3], [13], [16], [18], [19], [20], which suggest the EMG signals do not follow a steady Gaussian distribution even under constant force conditions. In other words, EMG variance can be a variable. Further, the sEMG variance can be affected by several factors, such as 1) physiological processes including motor unit recruitment, their firing rates, muscle fatigue, and individual differences in muscle anatomy and function [21]; 2) measurement factors including noise and movement artifacts [5]; 3) environmental conditions such as activity location and time of the day [22], [23]; and 4) personal conditions, such as age, experience, and body position, etc. [22], [24], [25]. A short survey on the impact of these factors is available in the Supplementary Material.

Addressing these factors with a deterministic model of variance of sEMG signals is challenging and impractical. However, a probabilistic model can simplify incorporation of the above contextual factors and allow a realistic representation of signal variations. Moreover, for applications such as prosthesis control, rehabilitation, and motion analysis, real-time estimation of EMG signal characteristics is crucial. Modeling variance as a random variable allows the system to quickly adapt to changes in the signal, ensuring accurate and responsive control [26]. Hence, the motivation for employing compound models for sEMG signals stems from the need to accurately capture their complex and nonstationary characteristics.

C. Review of CG Models for sEMG

Recently, the sEMG signals are modeled as nonstationary signals with heavy-tailed distributions; for example, Furui et al. [10] proposed a scale-mixture model (also known as a compound Gaussian (CG) model) to account for the nonstationarity of sEMG signals at different muscle contraction forces by considering variance as a hidden random variable that was assumed to follow an inverse gamma (IG) distribution. They have demonstrated a relationship between the variance distribution parameters and the muscle force. Furui et al. [26] developed an application of the scale mixture model in a pattern recognition task. It was shown that the scale mixture model-based classifier is superior to traditional discriminative models [26] based on classical EMG features such as root mean square (rms), the time domain statistics, and the frequency domain statistics. [27].

D. Contributions

In this study, the objective is to investigate nonstationary models for multivariate sEMG signals. For example, in [26], a multivariate compound model was proposed where the latent variable replaces the variance in univariate models and follows the IG distribution. However, the suitability of the IG distribution was not evaluated through comparison with other possible distributions commonly used in CG modeling. Some of the other possible models include gamma, exponential, and inverse Gaussian distributions [10], [28]. Hence, identification of a suitable distribution for the latent variable of the covariance that best fits the nonstationary sEMG signal characteristics is the focus of this study. The major contributions of this study are as follows. A CG model is proposed for nonstationary surface EMG signals with the latent variable of the covariance following an exponential distribution. A new dataset of sEMG signals corresponding to weight training exercises is developed and named Electromyography Analysis of Human Activities-Database 2 (EMAHA-DB2). The proposed model is tested on EMAHA-DB2, and its suitability is compared against the existing models using both the qualitative and quantitative approaches. Finally, the rate parameter of the proposed model and the multichannel signal power are analyzed for their dependencies on different measurement conditions.

The rest of this article is organized as follows. Section II presents the proposed model and its parameter estimation using the expectation maximization (EM) algorithm, followed by model validation methods. Section III describes the dataset. Section IV presents model analysis and discussion. Finally, Section V concludes this article.

II. Statistical Model and Problem Description

A. CG Model

1) Model Description: As mentioned earlier, the sEMG signals are considered nonstationary. Inspired by the scale mixture model for nonstationary EMG signals [10], an improved compound probabilistic model is proposed for the strength of multivariate sEMG signals. Specifically, the multichannel signal is modeled as the product of two interacting random variables. The first variable represents the fast-changing sEMG signal strength, while the second is a slow-varying latent random variable. Thus, proposed model for the multichannel sEMG samples $\mathbf{y}_{n,k}$ is as follows:

$$\mathbf{y}_{n,k} = \sqrt{z_k}\mathbf{x}_{n,k}. \tag{1}$$

Here, n denotes the sample index, and $\mathbf{x}_{n,k}$ represents the fast-changing random process modeled as a zero-mean multivariate Gaussian process within each kth segment, and z_k is the slow-changing hidden variable. The selection of z_k as a random variable modulating the underlying Gaussian component stems from the nonstationary nature observed in sEMG signals [10]. The underlying variations among the multiple channels are quantified by the spatial covariance structure in which z_k modulates the variance (in other words, z_k has an impact on each channel). In the univariate case, the variance in [10] and z_k are the same, whereas in the multivariate case, the z_k is associated with the spatial covariance and temporal fluctuations. Since z_k is modeled as a random variable, the variance of the sEMG signal of each channel is assumed to change for each k. Furthermore, it is assumed to be fixed within each kth interval. Here, $T = N \times K$ is the total number of samples in each channel, K denotes the number of segments in each channel, and N denotes the number of samples within each kth segment. Fig. 1 illustrates the relation between the two-channel sEMG signal and the compound statistical model. The pdf of $\mathbf{y}_{n,k}$ conditioned on z_k is defined as follows:

$$p(\mathbf{y}_{n,k}|z_k) = \frac{1}{(2\pi z_k)^{d/2}\,|\mathbf{\Sigma}|^{1/2}} \exp\left(-\frac{Q(\mathbf{y}_{n,k})}{2z_k}\right) \tag{2}$$

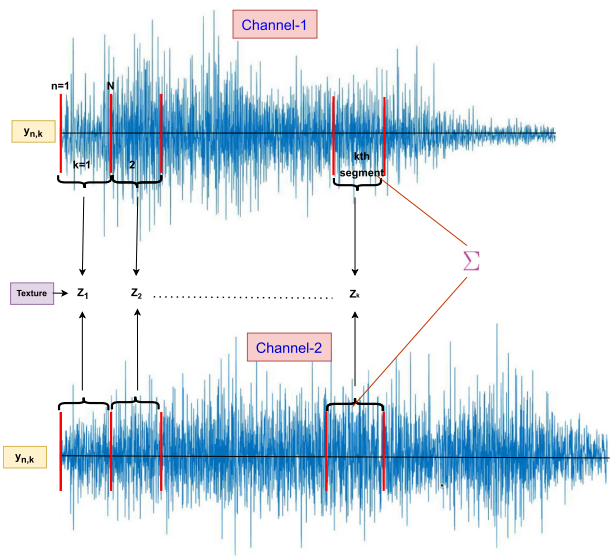

Fig. 1. Illustration of multivariate compound statistical for sEMG.

where the quadratic function

$$Q(\mathbf{y}_{n,k}) = (\mathbf{y}_{n,k})^T \mathbf{\Sigma}^{-1} (\mathbf{y}_{n,k}) \tag{3}$$

and $\mathbf{\Sigma}$ and d represent the spatial covariance matrix and the number of channels under consideration, respectively. In general, multichannel signals analyzed across K segments can have spatio-temporal correlations defined by the following spatio-temporal covariance matrix:

$$\mathbf{\Sigma}_{\text{ST}} = \mathbf{\Sigma}_T \otimes \mathbf{\Sigma} \tag{4}$$

where $\mathbf{\Sigma}_T$ represents the temporal correlations. However, for the sake of simplicity, the temporal correlations are reduced to a scalar (z_k). Hence, the conditional covariance of $\mathbf{y}_{n,k}$ reduces to

$$\mathbf{\Sigma}_{\text{ST}} = z_k \mathbf{\Sigma}. \tag{5}$$

Furthermore, it is assumed that the EMG signal decorrelates across the segments, a common phenomenon observed for signals modeled as CG. Note that in [10], a single channel sEMG signal was modeled, and the covariance was further reduced to a scalar variance modeled as an IG random variable.

To motivate the choice of probabilistic model for z_k, consider the process of sEMG signal generation. As the final link in the neural pathway to motor control, axons of motor neurons innervate the muscle fibers and determine the muscle activation. The innervation numbers are found to be exponentially distributed across a motor unit pool [29], [30], [31]. Further, the muscle fiber activation (recruitment of muscle units) is a time-varying process during muscle contraction and has a time-varying relation with MUAPs. The sEMG signals capture recruitment of these muscle fibers [32]. An existing challenge is to identify statistical properties of sEMG signals that relate to this underlying signal generation process. Hence in this study, the hidden variable z_k, modeled by an exponential distribution, is proposed to be related to muscle unit activation and subsequently to the innervation numbers.

Thus, the pdf of z_k is defined as follows:

$$p(z_k) = \frac{1}{\lambda} \exp\left(-\frac{z_k}{\lambda}\right). \tag{6}$$

Furui et al. [10] illustrated a relation between the parameters of the variance (IG distribution) and the muscle force. Similarly, in the context of an exponential distribution of z_k, λ refers to the rate parameter and plays a similar role [33]. It determines the statistical properties of the z_k. Specifically, it determines the mean and variance of z_k across the signal length. A higher value of λ indicates a higher mean and more spread-out distribution of z_k, reflecting a higher degree of muscle unit recruitment and activation variability. Consequently, understanding λ provides insights into underlying physiological mechanisms during different contractions and activity intensities. Based on the model in (6), the marginal distribution of $\mathbf{y}_{n,k}$ can be obtained by integrating out the hidden variable z_k as follows:

$$\begin{aligned} p(\mathbf{y}_{n,k}) &= \int_0^\infty p(\mathbf{y}_{n,k}|z_k) p(z_k) dz_k \\ &= \frac{1}{(2\pi)^{\frac{d}{2}} \lambda |\Sigma|^{\frac{1}{2}}} \int_0^\infty z_k^{\frac{-d}{2}} e^{-} \left(\frac{T_k^1}{z_k} + \frac{z_k}{\lambda}\right) dz_k \end{aligned} \tag{7}$$

where T_k^1 is defined as follows:

$$T_k^1 = \frac{1}{2} \sum_{n=1}^{N} Q(\mathbf{y}_{n,k}). \tag{8}$$

Using [34, ET II 82(23)a, LET I 146(29)] the following integral is identified:

$$\int_0^\infty x^{\vartheta-1} \exp\left(-\frac{A}{x} - Bx\right) dx = 2\left(\frac{A}{B}\right)^{\frac{\vartheta}{2}} K_\vartheta\left(2\sqrt{AB}\right) \tag{9}$$

where $K(\cdot)$ represents the modified Bessel function of second kind, ϑ is the order of Bessel function, and A and B are its parameters. Using (9) the marginal distribution (8) becomes [35] the following:

$$p(\mathbf{y}_{n,k}) = \frac{2}{(2\pi)^{\frac{d}{2}} |\Sigma|^{\frac{1}{2}} \lambda} \frac{K_{\frac{d}{2}-1}\left(\sqrt{\frac{2Q(y_{n,k})}{\lambda}}\right)}{\left(\sqrt{\frac{\lambda Q(y_{n,k})}{2}}\right)^{\frac{d}{2}-1}}. \tag{10}$$

2) Application of CG-E Model: The pdf of sEMG signals is important in the statistical classification of these signals. The model is specifically used in the process of feature extraction. There are multiple studies that used statistical models for discriminating human activities [26], [36], [37]. Likewise, the proposed model from this study is applied to categorize strength training status (STS) [38] of athletes. The estimates of parameters derived from this model are then employed as features for discriminating STS. In this context, the authors have conducted a comparative analysis of the classification performance against other statistical models, including state-of-the-art scale mixture models and other model features. It is observed that among these models, utilizing the parameters of the CG model as features yielded superior performance.

B. Estimation Problem

The complete data likelihood model can be written as follows:

$$p(\mathbf{Y}, \mathbf{z}; \boldsymbol{\mu}, \boldsymbol{\Sigma}, \lambda) = \prod_{k=1}^{K} \prod_{n=1}^{N} p(\mathbf{y}_{n,k}|z_k; \boldsymbol{\mu}, \boldsymbol{\Sigma}) p(z_k; \lambda) \quad (11)$$

where the full set of samples is as follows:

$$\mathbf{Y} = \{\mathbf{y}_k\}_{k=1}^{K} \quad (12)$$

and $\mathbf{y}_k = \{\mathbf{y}_{n,k}\}_{n=1}^{N}$ is the set of N samples within kth segment, the set of variables $\mathbf{z} = \{z_k\}_{k=1}^{K}$. The parameter set is $\Theta = \{\lambda, \boldsymbol{\Sigma}\}$ and assumed to be deterministic and unknown.

The estimation problem is summarized as follows. Given the full data likelihood function (11), a set of measurements $\mathbf{Y}$ that follows the conditional distribution in (2), the variables $\mathbf{z}$ assumed to follow the exponential model (6), the objective is to estimate the posterior distribution of $\mathbf{z}$ and the unknown parameters Θ and assess the estimation performance.

C. Parameter Estimation Using EM Algorithm

Regarding statistical models involving hidden variables (11), the model parameters are usually estimated using an iterative EM algorithm [39]. In this study, the z_k moments and the unknown parameters Θ are estimated using the EM algorithm described in [40] and variations of the EM for the CG-E are available in [35]. It involves the E-step and the M-step as described below.

1) E-Step: In this step, the posterior distribution of the z_k is evaluated for all k based on the logarithm of the complete data likelihood model (11), which is written as follows:

$$\begin{aligned} L &= \sum_{k=1}^{K} \sum_{n=1}^{N} \ln\left(p(y_{n,k}|z_k)\right) + \sum_{k=1}^{K} \ln\left(p(z_k)\right) \\ &= -\frac{NKd}{2}\ln(2\pi) - \frac{Nd}{2}\sum_{k=1}^{K}\ln(z_k) - \frac{NK}{2}\ln(|\boldsymbol{\Sigma}|) \\ &\quad - \frac{1}{2}\sum_{k=1}^{K}\sum_{n=1}^{N}\frac{Q(\mathbf{y}_{n,k})}{z_k} - K\ln(\lambda) - \frac{1}{\lambda}\sum_{k=1}^{K} z_k. \end{aligned} \quad (13)$$

As mentioned earlier (14), z_k are independent across segments; thus, at each iteration, the posterior pdf of $\mathbf{z}$ can be approximated as the product of individual posteriors as follows:

$$q_j(\mathbf{z}) = \prod_{k=1}^{K} q_j(z_k). \quad (14)$$

Gathering the z_k terms in (13), the log posterior of z_k is obtained as follows:

$$\ln(q(z_k)) = -\frac{Nd}{2}\ln(z_k) - \frac{T_k^1}{z_k} - \frac{z_k}{\lambda} + C. \quad (15)$$

Applying the exponential function on both the sides leads to the following:

$$q(z_k) \propto z_k^{-\frac{Nd}{2}} e^{-\frac{T_k^1}{z_k} - \frac{z_k}{\lambda}}. \quad (16)$$

The normalization constant for above pdf is evaluated as follows:

$$V = \int_0^{\infty} q(z_k) dz_k \quad (17)$$

using (9) V becomes

$$\begin{aligned} V &= \int_0^{\infty} z_k^{-\frac{Nd}{2}} e^{-\frac{T_k^1}{z_k} - \frac{z_k}{\lambda}} dz_k \\ &= 2\left(\frac{T_k^1}{1/\lambda}\right)^{\frac{\varsigma}{2}} K_{\varsigma}\left(2\sqrt{T_k^1/\lambda}\right) \end{aligned} \quad (18)$$

where ς is the order of the Bessel function and is defined as follows:

$$\varsigma = \frac{-Nd}{2} + 1. \quad (19)$$

The posterior distribution of z_k is given by the pdf as follows:

$$q(z_k) = \frac{z_k^{-\frac{Nd}{2}} e^{-\frac{T_k^1}{z_k} - \frac{z_k}{\lambda}}}{2\left(\frac{T_k^1}{1/\lambda}\right)^{\frac{\varsigma}{2}} K_{\varsigma}\left(2\sqrt{T_k^1/\lambda}\right)}. \quad (20)$$

The estimated posterior (20) is not a known distribution. However, expectations of functions $h(z_k)$ required in the following M-step can be evaluated as follows:

$$\langle h(z_k)\rangle = \int_0^{\infty} h(z_k) q(z_k) dz_k. \quad (21)$$

By substituting for $q(z_k)$ from (20) and utilizing (9), the posterior mean [41] is obtained as follows:

$$\langle z_k\rangle = \sqrt{\left(\frac{T_k^1}{1/\lambda}\right)} \frac{K_{\varsigma+1}(2\sqrt{T_k^1/\lambda})}{K_{\varsigma}(2\sqrt{T_k^1/\lambda})}. \quad (22)$$

Similarly, the other required moments $\langle \ln(z_k)\rangle$ and $\left\langle \frac{1}{z_k}\right\rangle$ are evaluated as follows:

$$\langle \ln z_k\rangle = \frac{1}{2}\ln\left(\frac{T_k^1}{1/\lambda}\right) + \frac{\left.\frac{\partial K_{\xi}(2\sqrt{T_k^1/\lambda})}{\partial \xi}\right|_{\xi=\varsigma}}{K_{\varsigma}(2\sqrt{T_k^1/\lambda})} \quad (23)$$

$$\left\langle \frac{1}{z_k}\right\rangle = \left(\frac{T_k^1}{1/\lambda}\right)^{-\frac{1}{2}} \frac{K_{\varsigma-1}(2\sqrt{T_k^1/\lambda})}{K_{\varsigma}(2\sqrt{T_k^1/\lambda})}. \quad (24)$$

2) M-Step: In this step, the moments from the E-step are utilized to compute the unknown parameters. The expectation of log likelihood function is as follows:

$$\begin{aligned} \langle \ln L\rangle_{q(z_k)} &= -\frac{Nd}{2}\sum_{k=1}^{K}\langle \ln z_k\rangle - \sum_{k=1}^{K}\left\langle \frac{1}{z_k}\right\rangle T_k^1 - \frac{NK}{2}\ln(|\boldsymbol{\Sigma}|) \\ &\quad - K\ln(\lambda) - \frac{1}{\lambda}\sum_{k=1}^{K}\langle z_k\rangle. \end{aligned} \quad (25)$$

The parameters are estimated by maximizing the expectation as follows:

$$\nabla_{\mu,\Sigma,\lambda}\langle \ln(L)\rangle_{q(z_k)} = 0. \quad (26)$$

Algorithm 1: EM-Algorithm for CG-E Model.

Input: Measurements Y
Initialize the parameters λ_0, Σ_0
Set $i = 1$
while $i \leq I_{\max}$ **do**
(a) **E-step:**
  for $j = 1 \rightarrow J$ **do**
Update:
    the statistic T_k^1 in (8)
    moments $\langle z_k \rangle$, $\left\langle \frac{1}{z_k} \right\rangle$ and $\langle \ln z_k \rangle$ in (22) to (24)
  end for
(b) **M-Step:**
    Update estimates $\lambda, \boldsymbol{\Sigma}$ in (27) and (28)
(c) **Stopping Criterion:**
    Update $\phi^{(i)}$ from (29)
      if condition in (30) is true
      Convergence reached, stop.
      else
      Set $i \leftarrow i + 1$
      Repeat steps (a), (b) and (c)
      end if
end while
Output: Estimated posterior distribution of z_k and ML estimates of λ, Σ

Using the moments from (22) to (24) in (26), the parameters are estimated as follows:

$$\hat{\lambda} = \frac{1}{K} \sum_{k=1}^{K} \langle z_k \rangle \tag{27}$$

$$\hat{\Sigma} = \frac{1}{NK} \sum_{k=1}^{K} \sum_{n=1}^{N} \left\langle \frac{1}{z_k} \right\rangle \mathbf{Y}\mathbf{Y}^{\intercal}. \tag{28}$$

Note that the estimates of λ and $\boldsymbol{\Sigma}$ are common for all k and they are estimated once in each iteration.

3) Convergence Criteria: The E and M steps are repeated until the convergence criterion defined below is satisfied. $\phi^{(i)}$ represents the sum of absolute change in the consecutive parameter estimates at the ith iteration and defined as follows:

$$\phi^{(i)} = \left|\lambda^i - \lambda^{i-1}\right| + \left|\Sigma^i - \Sigma^{i-1}\right|. \tag{29}$$

When the change $\phi^{(i)}$ becomes sufficiently small, i.e.,

$$\phi^{(i)} \leq \phi_o \tag{30}$$

the iterations are halted. Here, ϕ_o represents a predefined value of 10^{-5}. The EM algorithm is summarized in Algorithm 1.

D. Comparison Models

In this work, we compare our proposed model (CG-E) with the benchmark [26], compound Gaussian distribution with the inverse gamma hidden variable (CG-IG), and another model, compound Gaussian distribution with the gamma [28] hidden variable (CG-G).

1) CG-IG Model: The z_k follows an IG distribution

$$p(z_k) = \mathcal{IG}\left(z_k; \alpha_{\text{IG}}, \beta_{\text{IG}}\right) \tag{31}$$

and the conditional distribution of $y_{n,k}$ given z_k is the same as (2) with the parameter $\boldsymbol{\Sigma}_{\text{IG}}$. Note that this benchmark model is based on the scale mixture model in [10]. Let $\Theta_1 = \{\boldsymbol{\Sigma}_{\text{IG}}, \alpha_{\text{IG}}, \beta_{\text{IG}}\}$ be the parameter set for the CG-IG model. The EM algorithm for estimation of the parameters is summarized as follows, and similar results can be found in [10] and [26].

a) *E-step:* The posterior distribution of z_k has a closed-form expression, which follows the IG distribution, i.e.,

$$q(z_k) = \text{IG}\left(z_k; \alpha_{\text{IG}}^*, \beta_{\text{IG}}^*\right) \tag{32}$$

where α_{IG}^* and β_{IG}^* are written as follows:

$$\alpha_{\text{IG}}^* = \frac{Nd}{2} + \alpha_{\text{IG}}$$
$$\beta_{\text{IG}}^* = \beta_{\text{IG}} + T_k^1. \tag{33}$$

The moments of z_k are given as follows:

$$\langle \ln z_k \rangle = \ln(\beta_{\text{IG}}^*) - \psi(\alpha_{\text{IG}}^*)$$
$$\left\langle \frac{1}{z_k} \right\rangle = \frac{\alpha_{\text{IG}}^*}{\beta_{\text{IG}}^*}. \tag{34}$$

b) *M-step:* The estimates of Θ_1 are obtained as follows. An estimate of α_{IG} is found by solving the nonlinear equation

$$K\psi(\alpha_{\text{IG}}) - K\ln(\beta_{\text{IG}}) + \sum_{k=1}^{K} \langle \ln(z_k) \rangle = 0 \tag{35}$$

using the Newton–Raphson method [42] and by using the solution in (35), an estimate β_{IG} is obtained as follows:

$$\breve{\beta}_{\text{IG}} = \frac{K\breve{\alpha}_{\text{IG}}}{\sum_{k=1}^{K} \left\langle \frac{1}{z_k} \right\rangle}. \tag{36}$$

The estimate $\boldsymbol{\Sigma}_{\text{IG}}$ is similar to (28) except the moments of z_k are replaced by those in (34).

2) CG-G Model: Here, the z_k is considered as a gamma random variable.

$$p(z_k) = \mathcal{G}(z_k; \alpha_G, \beta_G). \tag{37}$$

The conditional distribution $p(y_{n,k}|z_k)$ is again similar to (2) with the set $\boldsymbol{\Sigma}_G$, α_G, β_G as the parameters of the gamma distribution. Let $\Theta_2 = \{\boldsymbol{\Sigma}_G, \alpha_G, \beta_G\}$ be the parameter set of the CG-G model. Note that this model is usually used for modeling nonstationary radar clutter [43].

a) *E-step:* The posterior distribution of z_k with the CG-G model is similar to (20) and is given as follows:

$$q(z_k) = \frac{z_k^{\alpha_G - \frac{Nd}{2} - 1} e^{-\frac{T_k^1}{z_k} - \beta_G z_k}}{2\left(\frac{T_k^1}{\beta_G}\right)^{\frac{\nu}{2}} K_\nu\left(2\sqrt{\beta_G T_k^1}\right)} \tag{38}$$

where $\nu = -\frac{Nd}{2} + \alpha_G$ represents the order of the Bessel function. The moments of z_k are similar to those of the CG-E model (22) to (24) with the following replacements:

$$\frac{1}{\lambda} \rightarrow \beta_G$$
$$\varsigma \rightarrow \nu. \tag{39}$$

b) *M-step:* The estimate of α_G is obtained by solving the nonlinear equation given as follows:

$$K\psi(\alpha_G) - K\ln(\beta_G) - \sum_{k=1}^{K} \langle \ln(z_k) \rangle = 0 \tag{40}$$

using this estimate of α_G, the estimate of β_G is obtained as follows:

$$\breve{\beta}_G = \frac{K\breve{\alpha}_G}{\sum_{k=1}^{K} \langle z_k \rangle}. \tag{41}$$

The estimate of $\mathbf{\Sigma}_G$ is similar to (28) except for the modified moments of z_k.

E. Evaluation Methods

Qualitative and multiple quantitative measures are presented to provide a multifaceted validation of the model's performance.

Visual inspection [44], [45]: It is a graphical approach to visualize the level of agreement between the histogram based empirical pdf (empdf) and an estimated pdf. In this study, these estimated compound pdfs are based on the CG-E, CG-G, and CG-IG models. To facilitate clearer comparisons, 1-D visualization plots are presented.

Moment analysis: In this analysis, the statistical moments estimated from the three models are compared with those of the empdf. For the three models, the closed-form joint pdfs lead to closed-form moments. The moments were evaluated as follows: For the CG-E, the covariance matrix is estimated from (28); for the CG-IG, it is evaluated using the above-mentioned estimator but by replacing $\left\langle \frac{1}{z_k} \right\rangle$ with the estimates from (34); and for the CG-G, the estimate is similar but replacements are from (39). For the empdf, the estimate is the sample covariance. The higher order statistics, Mardia's skewness and kurtosis are evaluated for each model and the empdf using (42) by substituting corresponding $\hat{\Sigma}$ from each of the above estimates, respectively, [46], [47].

$$\text{skew}_m = \frac{1}{N^2} \sum_{i=1}^{N} \sum_{j=1}^{N} \left[(Y_i)' \hat{\Sigma}^{-1} (Y_j) \right]^3$$

$$\text{kurt}_m = \frac{1}{N} \sum_{i=1}^{N} \left[(Y_i)' \hat{\Sigma}^{-1} (Y_i) \right]^2 \tag{42}$$

where Y and $\hat{\Sigma}$ are data and estimated covariance, respectively.

Kullback–Leibler divergence (KLD) [48]: It is a statistical metric used to measure the distance between two pdfs. Let q_1 and q_2 be the empdf and the estimated model pdf, respectively, then the KLD between them is evaluated as follows:

$$D_{\text{KL}}(q_1 || q_2) = \sum_x \sum_y q_1(x,y) \ln \left(\frac{q_1(x,y)}{q_2(x,y)} \right) \tag{43}$$

If these models match with each other then the $D_{\text{KL}}(q_1||q_2)$ equals 0. Thus, a lower $D_{\text{KL}}(q_1||q_2)$ indicates that an estimated model is closer to the empdf.

Coefficient of determination (COD) R-squared [49]: It measures the proportion of variance in the dependent variable that is predictable from the independent variables. Specifically, it quantifies how much of the overall variance the estimated model can explain. As the value of the R^2 approaches 1, the agreement between the estimated model and the empdf improves.

Log-Likelihood values (LLV) [44], [50]: Likelihood values indicate the probability that the given model would produce the observed data. Higher likelihood values suggest that the model's parameters are well-suited to the data. It is general and applicable to a wider range of statistical models, including nonlinear and complex signal models. We determine the LLV for the three models mentioned above.

Wilcoxon signed-rank test [51], [52]: It is a procedure used to compare two related samples or paired observations to determine whether their average ranks differ. It is particularly useful in situations where the data does not meet the normal distribution criteria required for parameteric tests such as the paired t-test. This makes it an excellent choice for dealing with ordinal data or when the data's distribution is unknown or skewed.

III. Data Description

In this work, a novel sEMG dataset termed EMAHA-DB2 is developed. Ten healthy participants aged between 18 and 21 years were selected based on three levels of strength training experience: a) Beginner with no prior training experience; b) Intermediate with a few weeks of training experience; and c) Trained with at least one year of training experience [53]. Participants were free from all muscle disorders for the past one month prior to data collection. Prior to participating in the experiment, the purpose of the study was explained, and informed consent was obtained from the subjects. The data collection procedure was approved by the institutional ethics committee of the Indian Institute of Information Technology Sri City (No. IIITS/EC/2022/01) dated September 19, 2022, as per the principles of the Declaration of Helsinki. Before the data acquisition session, the surface of the skin at the muscle site under consideration is cleaned with an alcohol-based wipe to reduce the skin impedance. In EMAHA-DB2, sEMG signals are acquired using the Noraxon's Ultium sensors. The electrode locations were identified using [54], [55]. These Ag/AgCL electrodes are rectangular with dimensions of 40×21 mm and interelectrode distance of 20 mm. As shown in Fig. 2(a), Ultium sensors are placed at two muscle sites. 1) Biceps brachii (BB) representing the upper arm activity and 2) Flexor carpi ulnaris (FCU) representing the forearm activity. Signal acquisition characteristics of the sensor are: 16-bit A/D; sampling rate: 2000 samples/s; cutoff frequency: 20–450 Hz. The weights used during the activity include 0, 1, 2.5, 5, 6, 9, and 10 kg. During measurements, the subject is in a standing position, and the weight is placed on a table at a convenient height. Each activity has three phases: 1) rest (10 s), 2) action (5 s), and 3) release (3 s) with a total duration of 18 s. Each activity is repeated nine times. In order to avoid muscle fatigue, subjects rest for two minutes between different activities. The anthropometric details of the participants are shown in Table I, and a summary of the characteristics is presented in Table II. The EMAHA-DB2 dataset is available here.[1]

[1]Data is available online at https://dx.doi.org/10.21227/0e4c-zc53.

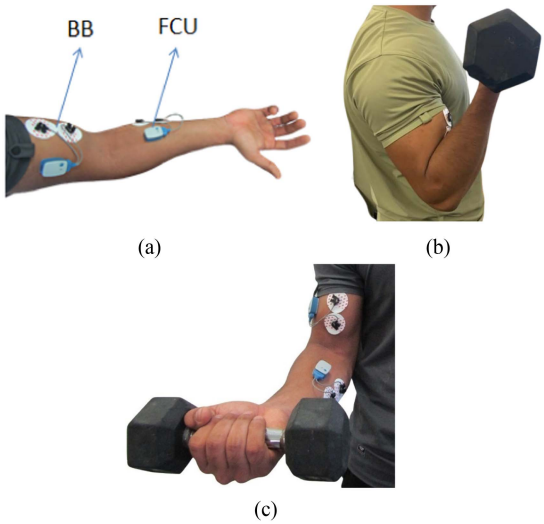

Fig. 2. (a) Placement of electrodes on the BB and FCU during weight training. (b) Isotonic activity: Performing bicep curls. (c) Isometric activity: Holding the dumbbell at 90°.

TABLE I
ANTHROPOMETRICS OF PARTICIPANTS

Subject	COBB*(in)	COFCU*(in)	Experience	Weight (kg)	Height (cms)
1	10.5	10	No	58	175
2	11.5	10.5	No	75	183
3	13	10.5	No	70	173.7
4	12.8	10.5	2 months	81	182
5	12.5	10	3 months	63	175
6	11	9.8	3 months	57	174
7	12	10	4 months	75	182
8	13.8	11.9	1 year	65	173
9	14.3	12	2 years	77	176
10	13.8	12.2	1 year	79	182.8

* COBB and COFCU stand for circumference of BB and FCU respectively.

TABLE II
CHARACTERISTICS OF EMAHA-DB2 DATASET

Weight	0kg to 10kg
Muscles	BB and FCU
Subjects	10
Rest duration	10 s
Activity duration	8 s
No of repetitions	09
sEMG sensor	Noraxon
Electrode	Agcl
Sampling frequency(Hz)	2000
No of channels	2

1) Experiment-I: In the first experiment, as shown in Fig. 2(b), the subjects were asked to perform bicep curls with the right arm using the seven weights mentioned above. Recall that the biceps curl corresponds to isotonic contractions [56].

2) Experiment-II: In this experiment, as shown in Fig. 2(c), the subjects were asked to hold a dumbbell with the right hand at 90° with respect to the upper arm, i.e., the dumbbell is held in the transverse plane with its axis parallel to the frontal axis. The same set of weights from Experiment I are used. Recall that for holding a weight, the arm flexion corresponds to isometric contractions [57].

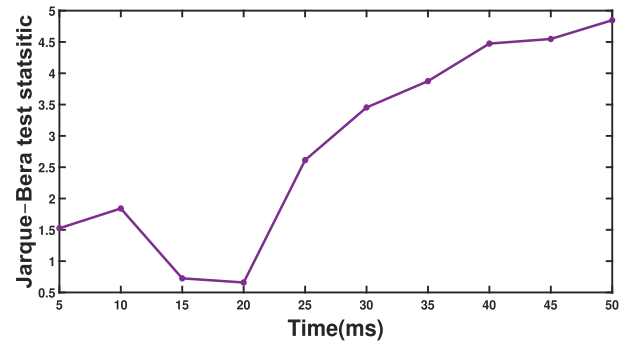

Fig. 3. Averaged JBT statistic across subjects and weights as a function of time.

IV. DATA ANALYSIS AND DISCUSSION

In this section, the most suitable model for the EMAHA-DB2 data is determined by comparing the following CG models:

1) CG-E (Proposed model);
2) CG-IG [10], [26];
3) CG-G [28].

Model validation is carried out for each of the sEMG signals corresponding to the experiments in Section III using the following evaluation methods:

1) Qualitative analysis based on visual inspection;
2) Quantitative analyses:
a) Moment analysis;
b) Analysis of KLD;
c) COD R^2;
d) LLV;
e) Wilcoxon statistical analyses.

The model analysis and the parameter estimation are carried out for optimal K and N. The Jarque–Bera test (JBT) is used to select the optimal values for K and N [58], [59]. For a given segment length, the test measures departure from Gaussianity. The point of minimal departure indicates the optimal segment length. The test statistic JBT is defined as $\text{JB} = \frac{n}{6}(\text{skew}^2 + \frac{1}{4}(\text{kurt} - 3)^2)$ where n is the variable segment length and skew and kurt are the sample skewness and kurtosis, respectively. The test is carried out for each activity of every subject considering the corresponding trials and weights. From Fig. 3, it is observed that on an average across the subjects and weights, the optimal choice of (K, N) equals (325,40).

A. Qualitative Analysis

1) Visual Inspection: Fig. 4 illustrates the empdf (yellow) and the pdfs from CG-E (blue), CG-G (magenta), and CG-IG (red) estimated for the strength of individual channels of sEMG signals. Specifically, Fig. 4(a) and (b) show the estimated pdfs of sEMG signals from the BB and FCU during isotonic contractions by subject-1 while training with a 6 kg dumbbell. Similarly, Fig. 4(c) and (d) are related to isometric contractions. From these plots, it is noticed that the CG-E model fits the empdf better in comparison to other models. The models CG-G and CG-IG are weaker fits compared to CG-E.

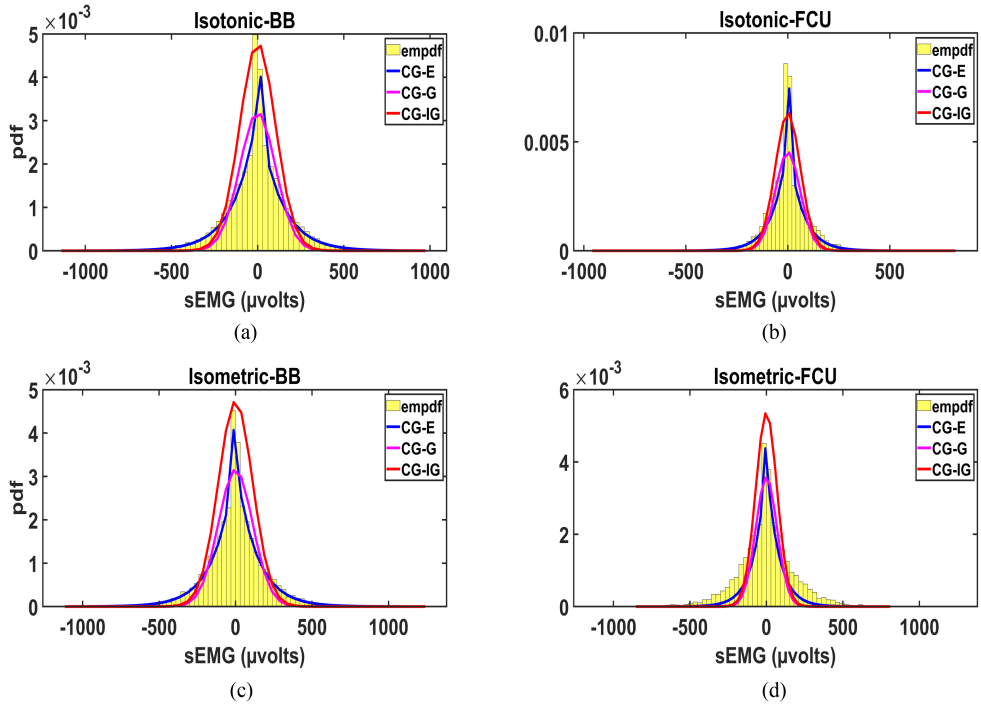

Fig. 4. Single channel visual comparisons between empdf (yellow) and CG-E (blue), CG-G (magenta) and CG-IG (red) models corresponding to (a) isotonic-BB, (b) isotonic-FCU, (c) isometric-BB, and (d) isometric-FCU by subject-1 during lifting 6 kg and trial-8.

TABLE III
AVERAGED ESTIMATED MOMENTS ACROSS ALL SUBJECTS AND TRIALS RELATED TO ISOTONIC ACTIVITY

Estimates	empdf	CG-E	CG-IG	CG-G
Covariance	$\begin{bmatrix} 6.2974 & 1.1273 \\ 1.1273 & 3.5187 \end{bmatrix} * 10^4$	$\begin{bmatrix} 6.0671 & 1.0914 \\ 1.0914 & 3.2768 \end{bmatrix} * 10^4$	$\begin{bmatrix} 7.4962 & 2.6174 \\ 2.6174 & 5.8213 \end{bmatrix} * 10^4$	$\begin{bmatrix} 5.7486 & 0.8564 \\ 0.8564 & 2.185 \end{bmatrix} * 10^4$
Mardia's Kurtosis	8.9541	7.4169	11.6847	5.8317

TABLE IV
AVERAGED ESTIMATED MOMENTS ACROSS ALL SUBJECTS AND TRIALS RELATED TO ISOMETRIC ACTIVITY

Estimates	empdf	CG-E	CG-IG	CG-G
Covariance	$\begin{bmatrix} 8.2897 & 3.4850 \\ 3.4850 & 5.6894 \end{bmatrix} * 10^4$	$\begin{bmatrix} 8.1574 & 3.3950 \\ 3.3950 & 5.4076 \end{bmatrix} * 10^4$	$\begin{bmatrix} 9.5147 & 3.1495 \\ 3.1495 & 4.6817 \end{bmatrix} * 10^4$	$\begin{bmatrix} 8.4147 & 2.5487 \\ 2.5487 & 4.8745 \end{bmatrix} * 10^4$
Mardia's Kurtosis	10.8274	9.5478	12.5814	6.4719

B. Quantitative Analysis

1) Moment Analysis: The average estimated moments such as covariance and Mardia's kurtosis [46], [47] across the subjects and trials for isotonic and isometric activities corresponding to 6 kg weight lifting are presented in Tables III and IV, respectively. Based on the covariance matrix, there exists a spatial correlation between the channels, indicating the channels are in coordination. Despite the nonstationary nature of the data, the covariance analysis helps in understanding the interchannel relationships and the coordination between different muscle groups. Note the estimated covariance component is assumed to be independent of time. The correlation between channels for isometric contractions ($\rho_{\text{isometric}} = 0.511$) is higher than for isotonic contractions ($\rho_{\text{isotonic}} = 0.2450$). These correlation estimates suggest that, compared to isotonic contractions, the isometric contractions have greater muscle coordination and effort. Further, the skewness is estimated to be approximately zero. Specifically, based on analysis of the distributions estimated from the EMG data, they are found to be symmetrical, which implies that both arms of the distribution are balanced, as illustrated in Fig. 4. Among the three models, the moments of CG-E are the best matches to those of the empdf. These results indicate agreement between the moments corresponding to the CG-E and the empdf.

2) KL-Divergence: In this study, the KLD is evaluated between the empdf and the three compound models and shown in Figs. 5 and 6. Fig. 5 illustrates KLD heatmaps as a function of subjects and activities. The KLD in each cell of the heatmap is an average over the trials of the corresponding activity. The KLDs corresponding to Experiment I are shown in Fig. 5(a)–(c), while the KLDs corresponding to Experiment-II are presented in Fig. 5(d)–(f). Based on these heatmaps, it is observed that the CG-E model has the lowest KLD among the three models. The

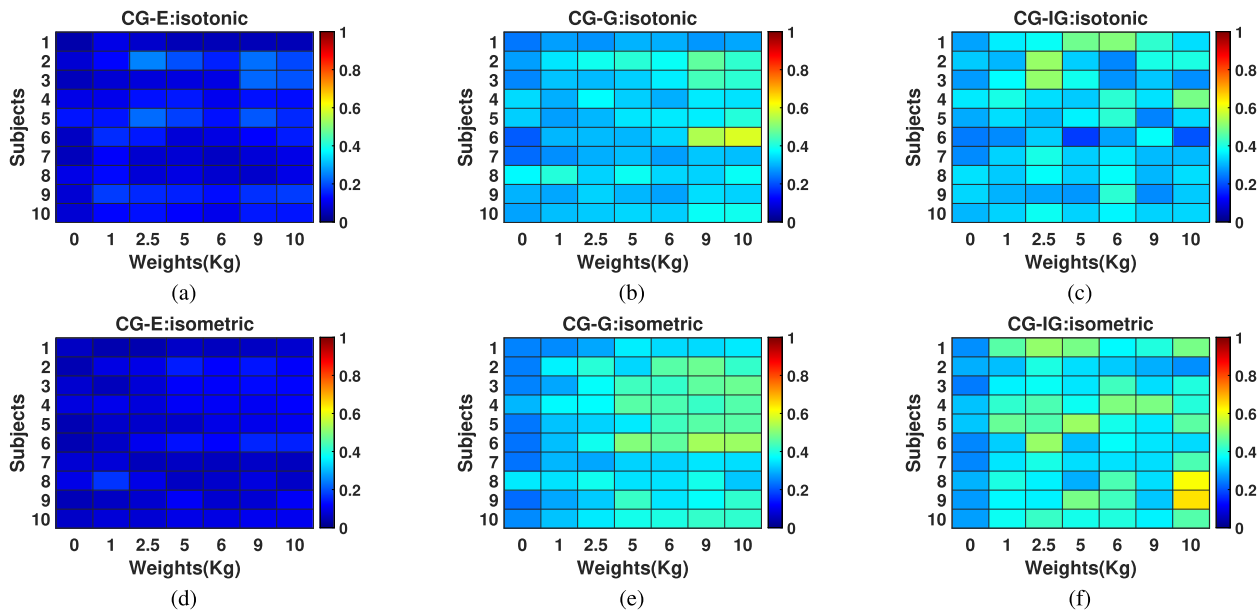

Fig. 5. Heatmaps of KLD: (a) CG-E, (b) CG-G, and (c) CG-IG from Experiment-I. (d) CG-E, (e) CG-G, and (f) CG-IG from Experiment-II.

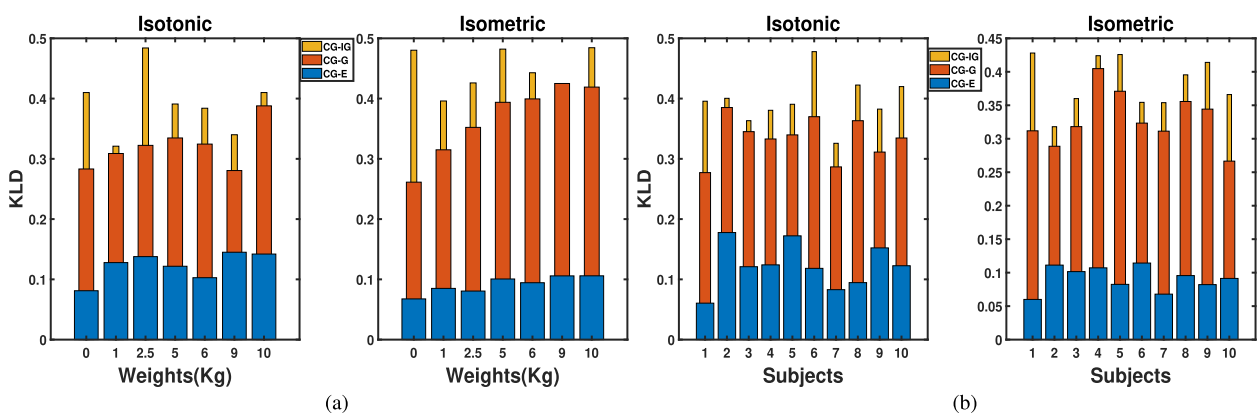

Fig. 6. KLD of CG-E, CG-G, and CG-IG for isometric and isotonic activities. (a) Averaged across subjects and plot versus weights. (b) Averaged across weights and plot versus subjects.

TABLE V
MINIMUM AND MAXIMUM VALUES OF KLD CORRESPONDING TO FIG. 5

Experiment	CG-E	CG-G	CG-IG
I	[0.0402 0.2506]	[0.214 0.5878]	[0.1817 0.5225]
II	[0.04487 0.1721]	[0.2286 0.5335]	[0.2428 0.65]

ranges of KLD for the heatmaps in Fig. 5 are given in Table V. For Experiment II, the maximum KLD from the CG-E does not exceed the minimum KLD from the CG-G and CG-IG. In the case of Experiment I, the maximum KLD from the CG-E is less than half the maximum from the other models.

Fig. 6(a) shows the averaged KLD across the subjects as a function of the weights, and Fig. 6(b) shows the vice versa. The KLDs of the CG-E, CG-G, and CG-IG are represented in blue, orange, and yellow, respectively. From the Fig. 6(a) and (b), it is noted that for both the experiments, the averaged KLD corresponding to either the subjects or the weights is the lowest for the CG-E when compared to CG-G and CG-IG.

3) Log-Likelihood Values: Fig. 7(a) illustrates the LLV averaged across the subjects and trials as a function of the weights. Fig. 7(b) shows the LLV averaged across the weights and trials as a function of the subjects. The LLV of the CG-E, CG-G, and CG-IG are shown again in blue, orange, and yellow, respectively. From these figures, it can be observed that for both the experiments, the averaged LLV for both the subjects and the weights is the highest for the CG-E among the three models.

4) COD R^2: The averaged R^2 for CG-E, CG-G, and CG-IG corresponding to Experiments I and II are illustrated in Fig. 8. Specifically, Fig. 8(a) shows R^2 averaged across subjects as a function of weights, and Fig. 8(b) shows vice versa. From these figures, it is clear that R^2 associated with the CG-E is the highest among the models, followed by that of the CG-G and the CG-IG. It is also evident that the difference in R^2 between CG-E and CG-G models is lesser in Experiment I; however, it is much greater with respect to Experiment II.

5) Statistical Analyses: Intermodel comparisons: To analyze the difference between the proposed CG-E model and the other

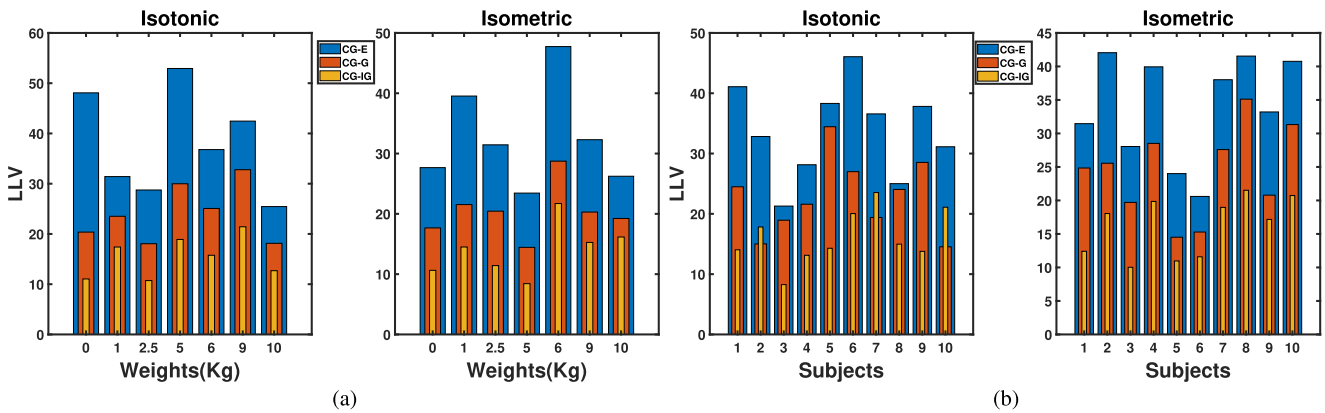

Fig. 7. LLV of CG-E, CG-G, and CG-IG for isometric and isotonic activities: (a) Averaged across subjects and plot versus weights. (b) Averaged across weights and plot versus subjects.

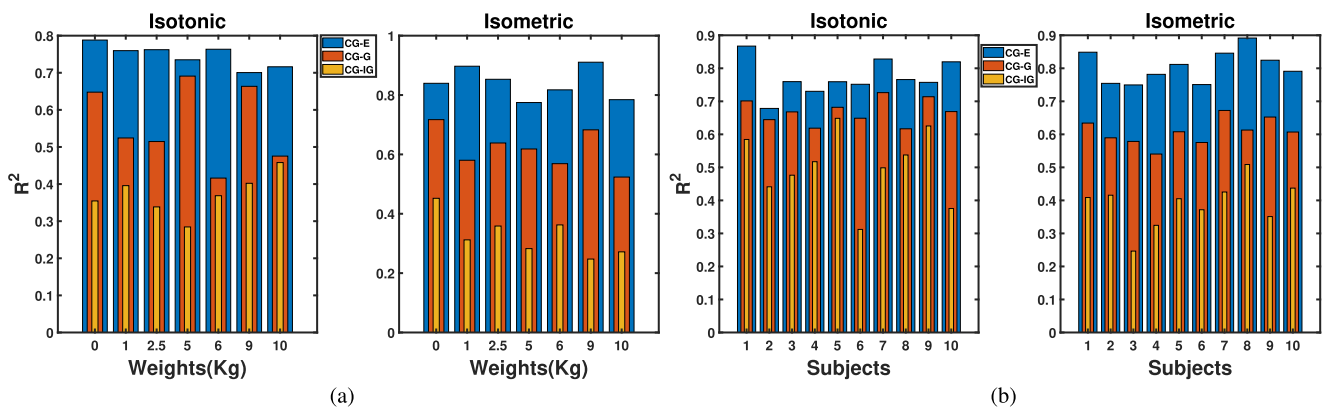

Fig. 8. R^2 of CG-E, CG-G, and CG-IG for isometric and isotonic activities: (a) Averaged across subjects and plot versus weights. (b) Averaged across weights and plot versus subjects.

TABLE VI
COMPARISON OF P-VALUES FOR DIFFERENT MODEL PAIRS

Model-1	Model-2	P_{Wilcoxon}	P_{Holm}
CG-E	CG-IG	$1.84 * 10^{-7}$	$4.57 * 10^{-31}$
CG-IG	CG-G	$9.45 * 10^{-4}$	$4.02 * 10^{-23}$
CG-G	CG-E	$2.91 * 10^{-6}$	$5.14 * 10^{-18}$

TABLE VII
COMPARISON OF P-VALUES BETWEEN THE EMPDF AND THE ESTIMATED PDFS FROM WILCOXON TEST

Model	P_{Wilcoxon}
CG-E	0.974
CG-IG	$1.509 * 10^{-47}$
CG-G	$2.972 * 10^{-28}$

(CG-IG and CG-G) models, statistical significance tests are performed. Specifically, the Wilcoxon signed rank test is performed pairwise for the three models under the null hypothesis (H_o): The models do not differ significantly. This test is carried out for 95% confidence interval. In order to maintain the overall familywise error rate (FWER) under a significance level, Holm's alpha correction is applied to the Wilcoxon test [60]. This method corrects Wilcoxon's p-values used in individual model comparisons. Table VI presents the pairwise model Wilcoxon p-values (p_{Wilcoxon}) and Holm's FWER p-values (p_{Holm}) for isometric activity with 6 kg by subject-1 during the trial-8. As can be seen, the null hypothesis can be rejected for each pair at significance level $\alpha = 0.05$. For the rest of the activities for different subjects and trials, similar p-values are observed. Hence, it can be inferred that the three models differ significantly from each other.

Comparisons with empdf: To analyze the agreement between the proposed models and the empdf, the Wilcoxon signed rank test [52] is performed between the empdf and the estimated pdfs under the null hypotheses (H_o): There is no difference between the empdf and estimated pdf. Table VII shows p-values of the Wilcoxon test for isometric activity with 6 kg by subject-1 during the trial-8. It is noticed that the p-value associated with CG-E is greater than 0.05, and those for CG-IG and CG-G are less than 0.05. The inference is that the null hypothesis is accepted for the CG-E model. Again, for different subjects and trials, similar p-values are observed for the tests on comparisons with empdf. This indicates that the CG-E model has superior agreement compared to other models.

C. Analysis of Parameter Estimates

1) Rate Parameter ($\hat{\lambda}$) of CG-E Model: Fig. 9 displays heatmaps illustrating the estimate of the rate parameter ($\hat{\lambda}$) as a function of weights (horizontally) and subjects (vertically)

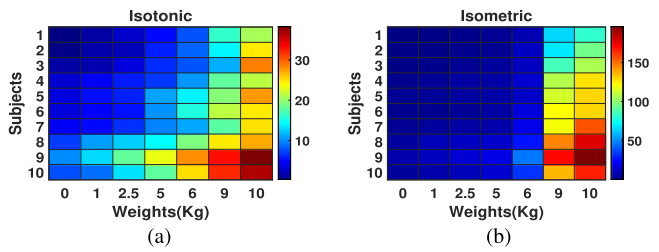

Fig. 9. Heatmaps for estimates of rate parameter $\hat{\lambda}$ averaged across trails: (a) Isotonic contraction. (b) Isometric contraction.

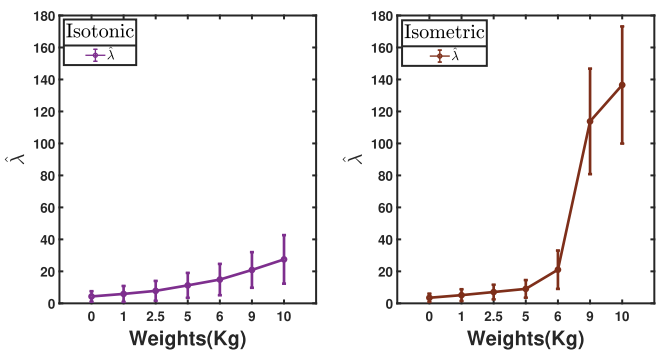

Fig. 10. Estimate of rate parameter ($\hat{\lambda}$) isotonic (left), isometric (right) as a function of weights averaged across trials and subjects. Error bars represent standard deviations for all subjects and trials.

averaged across the trials. Fig. 9(a) represents $\hat{\lambda}$ for isotonic contraction. Fig. 9(b) represents $\hat{\lambda}$ for isometric contraction. Fig. 10 depicts ($\hat{\lambda}$) as a function of dumbbell weights averaged across trials and subjects. First, it is noticed that as the dumbbell weight increases, there is a corresponding rise in $\hat{\lambda}$, suggesting a proportionality with dumbbell weights. In addition, note that the variation of $\hat{\lambda}$ is lower for Experiment I and higher for Experiment II. Furthermore, during the isometric contractions, the estimate $\hat{\lambda}$ increases only after 6 kg. However, it rises rapidly at higher weights of 9 and 10 kg. Note that the forces corresponding to isometric contractions are stronger than those in the isotonic contractions [61]. Furthermore, the large estimated lambda indicates that z_k may increase significantly over time. In terms of biomechanics, even though the force measurements are not available, one can definitely see a correlation between the $\hat{\lambda}$ and muscle force through the weight of the lifted dumbbell. Note that muscle force is related to the volume of motor units being recruited. This suggests that the $\hat{\lambda}$ might be indicative of volume of underlying muscle fiber activation. To ascertain whether there exists a disparity in $\hat{\lambda}$ of CG-E model between the experiments, the Wilcoxon test is conducted. As an illustration, the test is carried out on estimates of λ from the signals of the isometric and isotonic contractions corresponding to trial-8 with weight 6 kg from subject-1. The null hypothesis (H_o) is: There is no disparity in $\hat{\lambda}$ between the experiments. In this case, the obtained p-value is 0.00781, which is less than the significance level of $\alpha = 0.05$. Thus, there exists a statistically significant difference in $\hat{\lambda}$ between the experiments. To summarize, the key advantage of the probabilistic model of z_k over deterministic models is that it characterizes the variations of z_k at two levels.

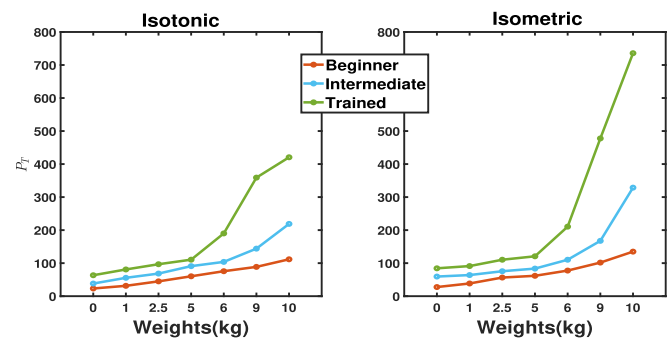

Fig. 11. Sum of estimated variances P_T as a function of weights averaged across trials and subjects.

One is understanding of z_k in terms of moment estimates at each interval. Second is capturing the overall behavior of z_k through λ over multiple intervals and thus for the entire activity and its relation with factors such as type of activity and weights used. This has no equivalent parameter in the deterministic case.

2) Analysis of Sum of Variances: The metric P_T denotes the square root of sum of variances (trace of Σ) from BB and FCU.

$$P_T = \sqrt{\sigma^2_{T,\text{BB}} + \sigma^2_{T,\text{FCU}}} \tag{44}$$

where $\sigma^2_{T,\text{BB}}$ and $\sigma^2_{T,\text{FCU}}$ are the sample variances corresponding to BB and FCU.

The P_T^2 termed *T-power* is the sum of variances of the sEMG signal from BB and FCU and can be related to the muscle force. Fig. 11 depicts P_T from BB and FCU for isotonic (left) and isometric (right) activities. From this figure, it is interesting to note that for both activities, for any lifting load, the total signal power seems to be directly related to the subject's training experience. For any weight, the trained subjects produced the highest T-power, while the beginners generated the least T-power. In addition, P_T can be correlated with a subject's strength. A higher slope for P_T versus weights indicates higher strength to lift heavier weights. Based on Fig. 11, the trained group has a steeper slope of P_T, followed by the intermediate and beginner groups.

In addition, for each of the ten subjects, a Wilcoxon test is conducted to assess the difference in P_T among the training categories. The null hypothesis (H_o) is: There is no significant disparity among the experience categories. It is noticed that all the p-values corresponding to training categories are less than the significance level of 0.05 ($p < 0.05$). Hence, the null hypothesis is rejected. It indicates that P_T values are different among experience categories.

In addition, standard deviations of P_T are shown in Fig. 12, with the left side corresponding to isotonic contractions and the right side to isometric contractions. In the first case, the variation in P_T appears to be linearly related to the weights. In the second case, the variations seem to increase at a faster rate with respect to weights, possibly an exponential rate. In both cases, the steady increase in variability in P_T can be attributed to the need for greater muscle unit recruitment at heavier weights. Further, at any dumbbell weight, isometric contractions may require greater recruitment in comparison to isotonic contractions.

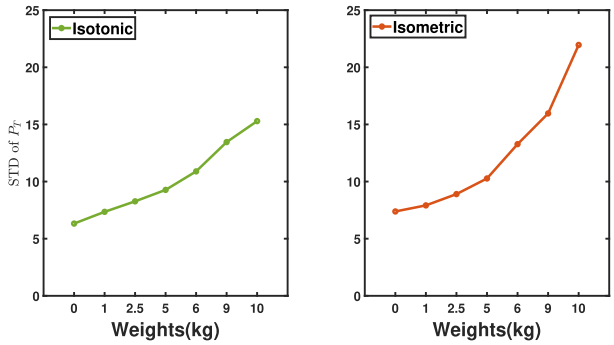

Fig. 12. Standard deviation of P_T as a function of weights averaged across trials and subjects.

V. Conclusion

In this article, a multivariate CG model is proposed for sEMG signals by considering an exponential model for the hidden variable. This model is compared with the existing (CG-IG) model. In addition to this, the proposed model is also compared with the CG model, in which the hidden variable is modeled by gamma distribution. The goodness of the model is justified using 1) a qualitative comparison with the empdf reveals the best agreement with the CG-E model, 2) comparison of the moments of the model pdf and the empdf, 3) the KLD between the fitted model and the empdf; again, the KLD is lowest for the CG-E model, 4) COD - R^2 is closest to unity in the case of the CG-E model, and 5) the LLV that also support the CG-E model. In addition, using the Wilcoxon rank test, it was shown that the CG-E model matches the empdf and it outperforms the other models. Furthermore, the estimates of $\hat{\lambda}$ between the experiments and P_T among the experience categories are statistically different. Finally, the estimates of the rate parameter ($\hat{\lambda}$) and the signal covariance of the (CG-E) model are analyzed in different measurement conditions. These findings suggest that $\hat{\lambda}$ might be indicative of underlying motor unit activity based on the dumbbell weights. In future work, the plans include statistical modeling of sEMG signals corresponding to sports activities and understanding their role in muscle coordination. Specifically, our focus will be on modeling and analyzing the temporal fluctuations in z_k using autoregression type models.

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